Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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10 Mathematical Induction

Prove the following statements with either induction, strong induction or proof by smallest counterexample.

2. For every integer $n \in \mathbb{N}$, it follows that $1^2 + 2^2 + 3^2 + 4^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1^2 = \dfrac{(1)(2)(3)}{6} = 1$, which is true.

2. Assume $1^2 + 2^2 + 3^2 + 4^2 + \ldots + k^2 = \dfrac{k(k+1)(2k+1)}{6}$ for some integer $k \geq 1$. Observe that this implies the statement is true for $n = k + 1$.

$$\begin{aligned}
&1^2 + 2^2 + 3^2 + \ldots + k^2 + (k+1)^2 \\
&\quad = (1^2 + 2^2 + 3^2 + \ldots + k^2) + (k+1)^2 \\
&\quad = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\
&\quad = \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \\
&\quad = \frac{k(k+1)(2k+1) + 6(k+1)(k+1)}{6} \\
&\quad = \frac{2k^3 + 9k^2 + 13k + 6}{6} \\
&\quad = \frac{(k+1)(k+2)(2k+3)}{6} \\
&\quad = \frac{(k+1)([k+1]+1)(2[k+1]+1)}{6}
\end{aligned}$$

Therefore $1^2 + 2^2 + 3^2 + \ldots + k^2 + (k+1)^2 = \dfrac{(k+1)([k+1]+1)(2[k+1]+1)}{6}$, which means the statement is true for $n = k+1$.

It follows by induction that the statement is true for every $n \in \mathbb{N}$. $\square$

4. If $n \in \mathbb{N}$, then $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + \ldots + n(n+1) = \dfrac{n(n+1)(n+2)}{3}$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, this statement is $1(1+1) = \dfrac{1(1+1)(1+2)}{3} = 2$, which is true.
2. Assume $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \ldots + k(k+1) = \dfrac{k(k+1)(k+2)}{3}$ for some integer $k \geq 1$. Observe that this implies the statement is true for $n = k+1$.

$$
\begin{aligned}
1 \cdot 2 + 2 \cdot 3 + \ldots &+ k(k+1) + (k+1)(k+2) \\
&= (1 \cdot 2 + 2 \cdot 3 + \ldots + k(k+1)) + (k+1)(k+2) \\
&= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\
&= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} \\
&= \frac{k(k^2+3k+2) + 3(k^2+3k+2)}{3} \\
&= \frac{k^3+6k^2+11k+6}{3} \\
&= \frac{(k+1)(k^2+5k+6)}{3} \\
&= \frac{(k+1)(k+2)(k+3)}{3} \\
&= \frac{(k+1)([k+1]+1)([k+1]+2)}{3}
\end{aligned}
$$

Therefore $1 \cdot 2 + 2 \cdot 3 + \ldots + k(k+1) + (k+1)(k+2) = \dfrac{(k+1)([k+1]+1)([k+1]+2)}{3}$, which means the statement is true for $n = k+1$.

It follows by induction that the statement is true for every $n \in \mathbb{N}$. $\square$

6. For every natural number n, it follows that $\sum_{i=1}^{n} (8i-5) = 4n^2 - n$.

Proof. We prove the statement by mathematical induction.

1. Observe that if $n = 1$, then the statement is $\sum_{i=1}^{1}(8i-5) = 4(1^2) - 1 = 3$, which is true.

2. Assume $\sum_{i=1}^{k}(8i-5) = 4k^2 - k$ for some $k \geq 1$. Then

$$\begin{aligned}
\sum_{i=1}^{k+1}(8i-5) &= \left[\sum_{i=1}^{k}(8i-5)\right] + \left[8(k+1)-5\right] \\
&= [4k^2 - k] + [8(k+1) - 5] \\
&= 4k^2 - k + 8k + 3 \\
&= 4k^2 + 8k + 4 - k - 1 \\
&= 4(k^2 + 2k + 1) - (k+1) \\
&= 4(k+1)(k+1) - (k+1) \\
&= 4(k+1)^2 - (k+1)
\end{aligned}$$

   This means the statement is true for $n = k + 1$.

It follows by induction that $\sum_{i=1}^{n}(8i-5) = 4n^2 - n$ for every natural number n. $\square$